Dissociable neural representations of reinforcement and belief prediction errors underlie strategic learning

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Decision-making in the presence of other competitive intelligent agents is fundamental for social and economic behavior. Such decisions require agents to behave strategically, where in addition to learning about the rewards and punishments available in the environment, they also need to anticipate and respond to actions of others competing for the same rewards. However, whereas we know much about strategic learning at both theoretical and behavioral levels, we know relatively little about the underlying neural mechanisms. Here, we show using a multi-strategy competitive learning paradigm that strategic choices can be characterized by extending the reinforcement learning (RL) framework to incorporate agents' beliefs about the actions of their opponents. Furthermore, using this characterization to generate putative internal values, we used model-based functional magnetic resonance imaging to investigate neural computations underlying strategic learning. We found that the distinct notions of prediction errors derived from our computational model are processed in a partially overlapping but distinct set of brain regions. Specifically, we found that the RL prediction error was correlated with activity in the ventral striatum. In contrast, activity in the ventral striatum, as well as the rostral anterior cingulate (rACC), was correlated with a previously uncharacterized belief-based prediction error. Furthermore, activity in rACC reflected individual differences in degree of engagement in belief learning. These results suggest a model of strategic behavior where learning arises from interaction of dissociable reinforcement and belief-based inputs.

game theory | neuroeconomics | computational modeling | functional MRI

Decision-making in the presence of competitive intelligent agents is fundamental for social and economic behavior (1, 2). Here, in addition to learning about rewards and punishments available in the environment, agents also need to anticipate and respond to actions of others competing for the same rewards. This ability to behave strategically has been the subject of intense study in theoretical biology and game theory (1, 2). However, whereas we know much about strategic learning at both theoretical and behavioral levels, we know relatively little about the underlying neural mechanisms. We studied neural computations underlying learning in a stylized but well-characterized setting of a population with many anonymously interacting agents and low probability of reencounter. This setting provides a natural model for situations such as commuters in traffic or bargaining in bazaars (1). Importantly, in minimizing the role of reputation and higher-order belief considerations, the population setting using a random matching protocol is perhaps the most widely studied experimental setting and has served as a basic building block for a number of models in evolutionary biology and game theory (1, 2).

Behaviorally, there is substantial evidence that strategic learning can be parsimoniously characterized by using two learning rules across a wide range of strategic contexts and experimental conditions: (*i*) reinforcement-based learning (RL) through trial and error, and (*ii*) belief-based learning through anticipating and responding to the actions of others (3, 4). The goal of this study is to provide a model-based account of the neural computations related to these two learning rules and their respective contributions to behavior. First, RL models have been central to understanding the neural systems underlying how reward learning (5). In the temporal-difference (TD) form, RL models posit that learning is driven by a prediction error defined as the difference between expected and received rewards and have been highly successful in connecting behavior to the underlying neurobiology (5, 6). Moreover, recent experiments in social and strategic domains have shown that RL models explain a number of important features of the data at both behavioral (3, 7) and neural levels (8, 9).

Despite their success, standard RL models provide an incomplete account of strategic learning even in the simple population setting. Organisms blindly exhibiting RL behavior in social and strategic settings are essentially ignoring that their behavior can be exploited by others (3, 10). In contrast, belief-based learning posits that players make use of knowledge of the structure of the game to update value estimates of available actions and comes in two computationally equivalent interpretations. One interpretation assumes the existence of latent beliefs and requires players to form and update first-order beliefs regarding the likelihood of future actions of opponents. Specifically, these models posit that players select actions strategically by best responding to their beliefs about future strategies of opponents and update these beliefs by using some weighted history of opponents' choices (1, 3). Mathematically, players engaging in belief learning correspond to Bayesian learners who believe opponent's play is drawn from a fixed but unknown distribution and whose prior beliefs take the form of a Dirichlet distribution (1). Under the alternative interpretation, beliefs and mental models are not assumed and action values are updated directly by reinforcing all actions proportional to their foregone (or fictive) rewards (11). The equivalence of these two mathematical interpretations thus makes it clear that belief-based learning does not necessarily imply the learning of mental, verbalizable beliefs commonly referred to in the cognitive and social sciences, because specific beliefs about likely strategies of opponents are sufficient but not necessary for this type of learning.

In this study, we used a multi-strategy competitive game, the so-called Patent Race (12), in conjunction with functional magnetic resonance imaging (fMRI). In the game, players of two types, Strong and Weak, are randomly matched at the beginning of each round and compete for a prize by choosing an investment (in integer amounts) from their respective endowments. The

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player who invests more wins the prize, and the other loses. In the event of a tie, both lose the prize. Regardless of the outcome, players lose the amount that they invested. In the particular payoff structure we use, the prize is worth 10 units, and the Strong (Weak) player is endowed with 5 (4) units (Fig. 1).

To illustrate how players can anticipate and respond to the actions of others in this game, suppose the Weak player observes the Strong players frequently investing five units. He may subsequently respond by playing zero to keep his initial endowment. Upon observing this play, Strong players can exploit the Weak player's behavior by investing only one unit to obtain both the prize while keeping four units from the endowment. This behavior may, in turn, entice the Weak player to move away from investing zero to win the prize. In contrast, pure RL players will respond to these changes in behavior of the opponents in a much slower manner, because they behave by comparing received payoffs from past investments without consideration for the strategic behavior of others (*SI Results* and Fig. S1).

This paradigm has three key features that build on insights from previous experimental and theoretical studies on learning models that together help to computationally characterize behavior and, statistically, minimize collinearity in the model outputs (13, 14). First, by using a random matching protocol, we minimize reputation concerns and, thus, the role of higher-order belief considerations (1). This paradigm allowed us to focus on first-order belief inferences, which are highly tractable and a key reason for its popularity in theoretical and experimental studies. Second, the large strategy space of the 6×5 game combined with the presence of secure strategies (i.e., investing zero or five), due to asymmetry in endowments between Strong and Weak players, allowed us to separate the relative contributions of belief and reinforcement inputs. The intuition is that secure strategies yield the same received payoffs regardless of actions by the opponents, giving us a control for the received payoff but still allowing the beliefs about the actions of the other players to change. In contrast, previous studies have typically used simple games that are well-suited to an experimental setting but statistically suboptimal in separating the relative behavioral contribution of two learning rules (13). Moreover, games with small strategy space often result in high negative correlation between foregone and received payoffs, making it problematic to dissociate the associated neural signals. Finally, to speed up game play, we replaced the standard matrix form display, which can be unintuitive even to highly educated subjects, with an interface that directly reflected the logic of the game (Fig. 1). In contrast, previous behavioral experiments yielding comparable behavior lasted over 2 h (*SI Results* and Table S1).

Results

Model Fits to Behavior. To characterize and disaggregate neural signatures of these two learning rules and their relationship to behavior, we adopted a hybrid model—experience weighted attraction—that combines and nests both reinforcement and belief learning (11, 15). The crucial insight connecting reinforcement and belief learning is to weight beliefs by using payoffs to obtain action values. That is, whereas action values are reinforced directly by the obtained outcomes in RL, in belief learning, they are weighted by beliefs that players hold about future actions of other players (*SI Methods*). This hybrid approach has been highly successful in explaining behavior across a wide range of games, thus offering a model-based framework to characterize the relative contributions of the two learning rules (3, 11).

Consistent with previous behavioral studies (3), the hybrid model outperformed both RL and belief learning models alone in explaining choices of subjects, as measured by the Bayesian Information Criterion penalizing for number of free parameters (Fig. 2*A*). In contrast, there was no significant difference between the base reinforcement and belief learning models. To account for overfitting, we conducted out-of-sample predictions by using holdback samples and found that the results were consistent with in-sample fits (Fig. 2*B*).

Separable Contribution of RL and Belief Inputs. Critically for our goal of disaggregating the neural signatures of the two learning rules, we next investigated whether behavior in the Patent Race was driven by subjects engaging in both reinforcement and belief inputs at the individual level, rather than a mixture of distinct types of pure reinforcement and belief learners. Using the hybrid model parameter ∂_i that governs the weighting between the two learning rules, we found that the individual estimates were distributed along the unit interval, rather than clustered at the boundaries as would be expected if the population consisted of distinct types (Fig. 2*C* and Table S2). This variation further allowed us to use these estimates as a between-subject measure in subsequent neuroimaging analysis. To test the robustness of our estimates to assumptions of the hybrid model, we used the log-likelihood ratio between RL and belief learning models and found that this measure was significantly correlated with ∂_i (Pearson $\rho = 0.70$, $P < 0.01$, two-tailed; Fig. 2*C*). This analysis can be interpreted as a model-free check that our individual difference measure was not unduly driven by assumptions underlying the hybrid model.

To illustrate the separable contributions of the respective learning rules to behavior, we compared the empirical choice frequencies of a single subject in the Weak role exhibiting both

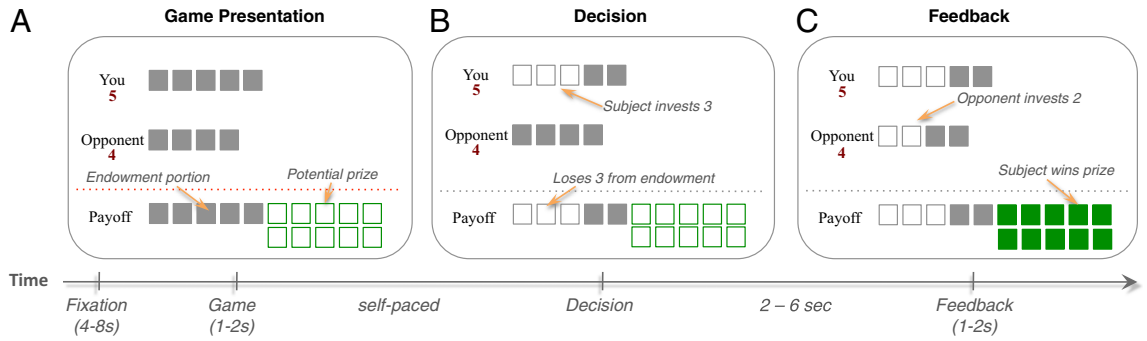


Fig. 1. Patent Race game. (*A*) After a fixation screen of a random duration between 4–8 s, subjects were presented with the Patent Race game for between 1–2 s, with information regarding their endowment, the endowment of the opponent, and the potential prize. (*B*) Subjects inputted the decision (self-paced) by pressing a button mapped to the desired investment amount from the initial endowment. (*C*) After 2–6 s, the opponent's choice was revealed. If the subject's investment is strictly more than those of the opponent, the subject won the prize; otherwise, the subject lost the prize. In either case, the subject kept the portion of the endowment not invested.

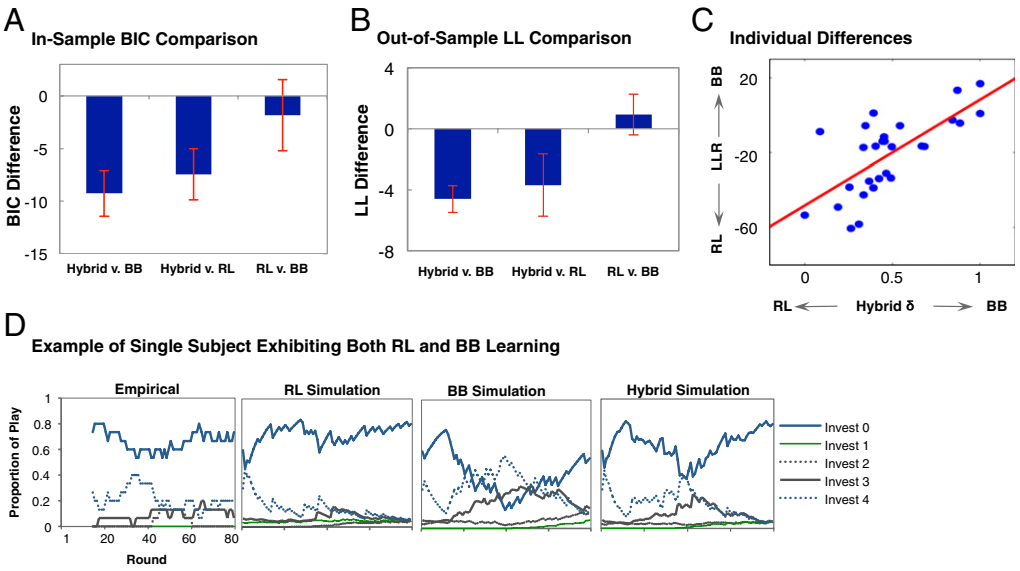


Fig. 2. Computational model estimates and single subject exhibiting both RL and belief learning. (*A*) In-sample model fit comparisons using the Bayesian Information Criterion showed that the hybrid model fits behavioral choices significantly better than RL and belief learning (paired Student's *t* test, $P \leq 0.01$, two-tailed), whereas RL and belief learning did not differ significantly among themselves. Error bars indicate SEM. (*B*) Out-of-sample predictive power was also superior for hybrid model compared with RL and belief learning (paired Student's *t* test $P \leq 0.01$, two-tailed). Error bars indicate SEM of log-likelihood differences. (*C*) Individual variation in the relative weights placed on RL and belief learning can be captured by using parameter δ_i of the hybrid model. As δ_i increases, behavioral fit of belief learning improves relative to that of the RL (Pearson $\rho = 0.70$, $P < 0.01$, two-tailed). (*D*) Illustration of behavior and model predictions using a single subject in the Weak role exhibiting both RL and BB learning. Empirical time series of choice is plotted by using a 15-round bin average. Choice probabilities were generated from calibrated models by using RL, belief, and hybrid learning models, respectively.

types of learning ($\partial_i = 0.47$) to simulations using the respective models (Fig. 2*D*). RL missed the increased probability of investing 4 in rounds 30–50. This corresponded to periods when Strong players invested one to three units with a high probability. In contrast, belief learning captured this change but overestimated its magnitude. By combining the two learning rules, the hybrid model was able to capture both the direction and magnitude of changes in investments reasonably well.

Ventral Striatum Activity Correlated with Both RL and Belief-Based Prediction Errors. Having characterized behavior of subjects computationally, we next sought to identify the brain regions where neural activity was significant correlated with the internal signals of each model. At the time of outcome, according to the TD form of the hybrid model, players update their action values by using a combination of the RL and belief components (*SI Methods*). Critically, the correlation between the two types of prediction errors is sufficiently low (Pearson $\rho \simeq 0.28$; Table S3) for us to characterize the unique contribution of the underlying neural signals.

First, we found that the RL prediction error was significantly correlated with activity in bilateral putamen and a small region in the cerebellum (Fig. 3*A* and Table 1). The RL prediction error is defined as the difference between expected and received rewards, and the striatum has been consistently implicated in encoding this quantity in fMRI studies of reward learning (16, 17). In contrast, we know much less about the neural underpinnings of belief learning. Studies of mentalization in social neuroscience have suggested that medial prefrontal cortices (mPFC) and temporo-parietal junction (TPJ) mediate social cognitive processes such as belief inference (18). However, studies on reward-guided behavior in social settings have suggested that these computations build on the same brain structures associated with reward, including the striatum, rostral anterior cingulate (rACC), and orbitofrontal cortex (19). Thus, using our model-based framework, we next sought to localize neural signatures associated with belief learning.

Surprisingly, we found that the belief prediction error for the chosen action was also correlated with activity in the putamen, but also extending to parts of the ventral caudate (Fig. 3*B*). To assess the robustness of this result to the correlation between the regressors, we searched for brain regions correlated with the

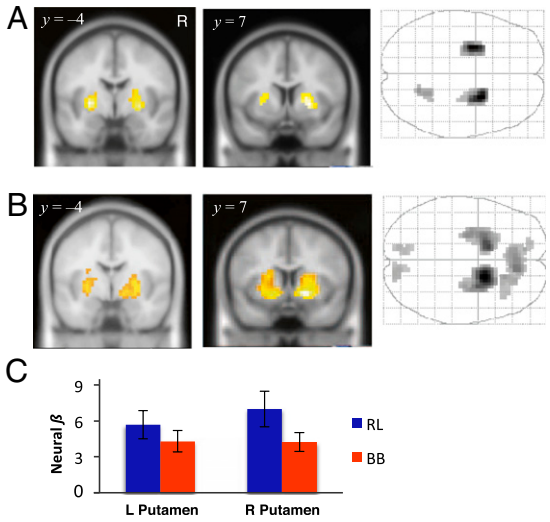


Fig. 3. Ventral striatal responses to RL and belief prediction errors. (*A*) Coronal sections and glass brain of putamen activation to the RL prediction error ($P < 0.001$ uncorrected, cluster size $k \geq 20$). (*B*) Significant activation of bilateral putamen and ventral caudate to the belief prediction error for the chosen action ($P < 0.001$ uncorrected, $k \geq 20$). (*C*) Region of interest analyses show that activities in bilateral ventral striatum (Table 1) were significantly correlated with both RL and belief prediction errors; error bars indicate SEM.

Table 1. List of brain activations responding to RL, belief, and hybrid prediction errors

Model	Region	Cluster level		Voxel level*				
		p_{cor}	Voxels	p_{fdr}	*T-val*	*X*	*Y*	*Z*
Reinforcement	R ventral striatum	0	120	0.004	6.01	24	4	0
	L ventral striatum	0.001	91	0.004	5.87	−28	−4	0
	R cerebellum	0.093	30	0.025	4.19	18	−60	−28
Belief-based	R ventral striatum	0	334	0	7.22	18	7	−7
	L ventral striatum	0	294	0	5.9	−14	10	−10
	rACC/mPFC	0	224	0.001	4.96	7	38	7
	R superior frontal gyrus	0.023	51	0.005	4.2	18	28	49
	R occipital cortex	0.057	39	0.009	3.9	14	−88	14
	L occipital cortex	0.107	31	0.013	3.72	−14	−84	7
Hybrid	R putamen	0	234	0	8.51	14	7	−7
	L putamen	0	148	0.004	6.91	−14	7	−10
	rACC/mPFC	0	150	0.18	5.3	7	38	35
	Occipital cortex	0	347	0.172	5.32	7	−84	4
	Cerebellum	0.111	28	0.006	4.64	−4	−56	−38

Regions significantly correlated with the different notions of prediction errors derived from the three models considered in this study. All activations survived a threshold of $P < 0.001$, uncorrected, and cluster size $k \geq 20$. L, left; R, right.
*Voxel locations given in MNI Coordinates.

unique share of variance attributed to RL and belief prediction errors, respectively. That is, we simultaneously included both the RL prediction error and the orthogonalized belief prediction error, and we verified that the striatal activation remained significant using this procedure (Fig. S2). The same procedure was used in reverse as a robustness check on the striatal activation to the RL prediction error.

Rostral ACC Activity Uniquely Correlated with Belief-Based Prediction Error. In contrast, we found that activity in the rACC extending to the mPFC was correlated with only the belief prediction error ($P < 0.001$, uncorrected, cluster size $k \geq 20$; Fig. 4*A* and Table 1), and not with RL prediction error even at a liberal threshold of $P > 0.5$. To verify the functional selectivity of the belief prediction error in the rACC, we conducted a paired Student's *t* test on the average beta values of the rACC activation and found that betas associated with belief prediction error were significantly greater than those for the RL prediction error ($P < 0.05$, two-tailed, Fig. 4*B*).

To investigate modulation of the two learning rules, we examined between-subject variability in the weighting placed on belief learning inputs by correlating the estimated behavioral parameter ∂_i to the neural response to the belief prediction error. Using brain regions that were found significantly correlated with the belief prediction error as regions of interest, we found that ∂_i was correlated with individual differences in activation of rACC (Pearson $P = 0.66$, $P < 0.01$, two-tailed; Fig. 4*C*), but not in the striatum ($P > 0.5$). That is, we found subject who assigned higher weights to belief learning behaviorally exhibited greater neural sensitivity in the rACC to the belief prediction error.

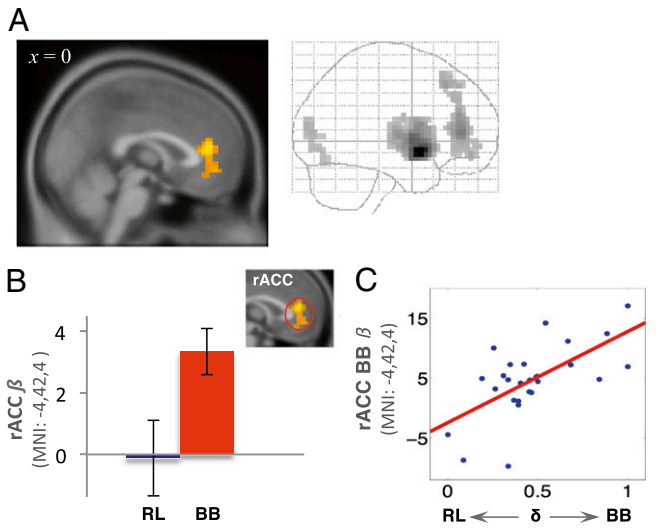


Fig. 4. Rostral ACC responds uniquely to belief prediction error. (*A*) Sagittal section and glass brain of rACC activation to the belief prediction error associated with the chosen action ($P < 0.001$ uncorrected, cluster size $k \geq 20$). (*B*) Neural activity in the rACC is correlated only with belief and not with RL prediction error, error bars indicate SEM. (*C*) Between-subject neural response to the belief prediction error in rACC is correlated with individual differences in behavioral engagement of belief learning as measured using ∂_i estimates (Pearson $P = 0.66$, $P < 0.01$, two-tailed).

fMRI Correlates of Hybrid Model. Finally, to test our hypothesis that the neural correlates of hybrid prediction error can be disaggregated into RL and belief-based components, we searched for brain regions significantly correlated with the hybrid prediction error. We found that the activations to the hybrid prediction error were contained by the union of activations associated with the RL and belief prediction errors, in particular the ventral striatum and the mPFC, and also parts of the occipital cortex and cerebellum (Table 1 and Fig. 5). In contrast, we found that reward predictions during the response period for all three models were represented in overlapping areas of the ventromedial PFC (Fig. S3). This finding thus suggests that inputs converge at the time of choice and is consistent with findings in animal literature of hybrid-type signals in sensory-motor regions (20).

Discussion

Notions of equilibrium, such as the well-known Nash equilibrium and related notions such as quantal-response equilibrium (21), are central to theories of strategic behavior. It has long been recognized, however, that equilibria do not emerge spontaneously, but rather through some adaptive process whereby organisms evolve or learn over time, which cannot be accounted for by purely static equilibrium models (3). Here, we studied the neural correlates of the adaptive process by using a unique multi-strategy competitive game. Our results show how the brain

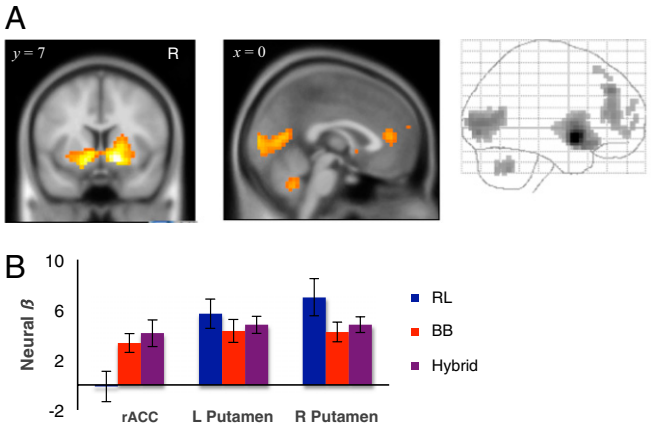


Fig. 5. Hybrid model prediction error activations. (*A*) Glass brain and section of bilateral putamen rACC activation in response to hybrid model prediction error ($P < 0.001$ uncorrected, cluster size $k \geq 20$). (*B*) Neural betas with respect to RL, belief, and hybrid prediction errors at three ROIs defined by 8-mm sphere at the peak of hybrid model activation clusters (Table 1).

responds to multiple notions of prediction errors derived from well-established behavioral models of strategic learning in a partially overlapping but distinct set of brain regions known to be involved in reward and social behavior.

First, we found that activity in the ventral striatum was correlated with both the RL and belief prediction errors. Unlike the RL prediction error, which has been extensively studied (16, 17), the belief prediction error corresponds to the discrepancy between the observed and expected frequency of actions of an opponent weighted by the associated payoffs. Thus, unlike RL models, belief learning predicts that players will be sensitive to foregone (or fictive) payoffs in addition to the received payoffs. Various notions of foregone payoffs have been studied in neuroimaging experiments of decision-making and reward learning, including fictive error, regret, and counterfactuals (22–24). Unlike these measures, our belief learning prediction error does not involve direct comparison of the possible payoffs, but rather requires subjects to incorporate their knowledge of the structure of the game to track payoffs associated with all actions to update their value estimates.

The overlapping set of activations in ventral striatum can potentially shed light on previous neuroimaging studies of social exchange games. These studies have consistently found neural signals that qualitatively resemble prediction errors in the striatum (25, 26). Our results raise several interesting possibilities regarding contents of the neural encoding. First, these regions may, in fact, contain, or receive information from, neurons whose activity represents multiple notions of prediction errors, including those related to belief-based considerations we study here. Alternatively, information processing in these regions may be sensitive to, or correlated with, beliefs and/or foregone payoffs, without necessarily encoding a literal belief prediction error. Future research will be needed to unravel the precise nature of the information processing in these regions.

In contrast to the ventral striatum, we found that activity in the rACC responded to the belief, but not the RL, prediction error. This finding is consistent with known roles of rostral and caudal ACC for error (27, 28) and conflict processing (29, 30), and in the social domain, mentalization and belief inference (18). When activation involves more anterior and rostral portions of the ACC, as in our case, it is usually in tasks with an emotional component (31, 32). Therefore, it seems a good candidate to modulate error processing and learning when it has a higher level, social, or regret based flavor (33). Furthermore, and unlike the putamen involvement in belief prediction error, rACC activity reflected individual differences in the sensitivity to belief inputs (Fig. 5*B*). Thus, although it is an open question precisely how these prediction errors interact at the trial level, these results suggest the role of rACC as a control region that mediates the degree of influence played by belief learning.

This distributed representation of different forms of prediction errors is consistent with other recent computational accounts of social behavior distinguishing between reward and social learning inputs. In a reward learning task where participants received social advice, activity in dorsomedial PFC was correlated with prediction errors associated with confederate's fidelity, whereas activity in the ventral striatum was correlated with the standard reward prediction error (34). A similar dissociation was found in an observational learning setting between the dorsolateral PFC and ventral striatum in encoding of observational and reward prediction errors, respectively (35). Unlike these studies, however, we found that activity in the ventral striatum contained both RL and belief updating signals. Although speculative, we conjecture that these differences may in part reflect differences in the cognitive demands for behaving in strategic settings versus more passive types of social learning, where subject's payoffs are not directly affected by the action of the other agents. Such a hypothesis can be tested by using studies that manipulate dependence of payoffs in strategic interactions.

Moreover, our results complement and extend current understandings of the role of the rACC and mPFC in mentalization in economic games. The rACC, but not the mPFC, was shown to be activated in subjects exhibiting first-level reasoning in an iterative reasoning game, whereas the reverse was true for subjects exhibiting second-level reasoning (36). Similarly, mPFC but not the rACC was implicated in higher-order mentalization in a competitive game (10). Together with our findings, these results raise the interesting possibility that more ventral mPFC regions, including rACC, are involved in first-order belief inference, and more dorsal regions in higher-order inferences. Such a functional division is consistent with previous proposals in the mentalization literature (18) and may reflect differences in the type of information being learned. That is, in addition to responding to history of actions of opponents, players engaging in higher-order mentalization may also respond to beliefs about the learning parameters of opponents (10). Future studies combining computational models of higher-order mentalization with games that directly manipulate the level of strategic reasoning of the players will be necessary to test this hypothesis.

Strategic learning is ubiquitous in social and economic interactions, where outcomes are determined by joint actions of the players involved, and is in sharp distinction to more passive social learning environments where social information could be manipulated independently of outcomes. The inherent correlation between actions and payoffs in strategic environments, however, introduces important methodological hurdles in disentangling the contributions of the underlying learning rules at both behavioral and neural levels. We were able to characterize them only through a tight integration of experiment design, computational modeling, and functional neuroimaging. More broadly, our results add to the growing literature suggesting that neural systems underlying social and strategic interactions can be studied in a unified computational framework with those of reward learning (19, 37). Although the current study is restricted to a specific competitive game, there is a wealth of theoretical and behavioral studies that suggest these learning rules generalize to other strategic settings, including not only competitive games but also those involving cooperation, iterative reasoning, or randomization (3). Such a general framework opens the door to understanding the striking array of social and behavioral deficits that appear to have their basis in damage and dysfunction related

to the same dopaminergic and frontal-subcortical circuits implicated in our study (38).

Methods

Participants. Thirty-five healthy volunteers (19 female) were recruited from Neuroeconomics Lab subject pool at the University of Illinois at Urbana–Champaign (UIUC). Subjects had a mean age of 23.3 ± 4.6 y, ranging 19–47. Of these subjects, five were excluded from the study because of excessive motion and three because of repeating the same strategy for >95% of the trials during the experiment.

Procedure. Subjects undergoing neuroimaging completed 160 rounds of the Patent Race game (Fig. 1) in two scanning sessions lasting 15–20 min each, alternating between Strong and Weak roles over 80 rounds, counterbalanced (*SI Methods*). Informed consent was obtained as approved by the Internal Review Board at UIUC. They were informed that they would be paid the average payoff from a randomly chosen 40 rounds from each session plus a $10 participation fee.

Behavioral Data Analysis. Details of the computational models used in the analysis are provided in *SI Methods*. All models learned the value of actions in a temporal-difference algorithm by tracking expected value of each action. The values differed only in the type of information that was used to form and update these expected values. For each model, we estimated by maximizing, for each subject, the log-likelihood of predicted decision probabilities generated by the models against the actual choices of subjects (*SI Methods*).

fMRI Data Analysis. Details of the fMRI acquisition and analysis are provided in *SI Methods*. Event-related analyses of fMRI time series were performed, with reward prediction and prediction errors values generated from the respective computational models. The decision event was associated with choice probabilities, and the feedback event was associated with prediction errors for chosen actions. All analyses were performed on the feedback event data, except the expected reward region analysis (Fig. S3). Regressors were convolved with the canonical hemodynamic response function and entered into a regression analysis against each subject's BOLD response data. The regression fits of each computational signal from each individual subject were then summed across their roles and then taken into random-effects group analysis (*SI Methods*).

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Supporting Information

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SI Methods

Procedure. Before entering the scanner, subjects were given instructions and completed a quiz to ensure comprehension of the game. In the Patent Race, players were matched at random at the beginning of each round and competed for a prize by choosing an investment from their respective endowments. The player who invested more won the prize, and the other lost. In the event of a tie, both lost the prize. Regardless of the outcome, players lost the amount that they invested (Fig. 1). In the particular payoff structure we used, the prize was worth 10 units, and the Strong (Weak) player was endowed with 5 (4) units.

To overcome logistic difficulties of conducting simultaneous experiments with upwards of 16 subjects for each neuroimaging subject, and to minimize unobserved session effects in opponent play associated with such a protocol, we matched subjects with choices from a pool of players who previously participated in behavioral sessions. Importantly, subjects were informed that they played in the same sequence as the pool players. That is, if the scanner subject was playing in round 60, the choice of opponent was drawn randomly from round 60 of one of the pool players (*SI Results*).

fMRI Scanning Parameters. Functional MR images were obtained for each subject by using a 3.0 Tesla Siemens Allegra scanner located at the research-dedicated Beckman Imaging Center (BIC) at the University of Illinois at Urbana–Champaign. Images were acquired by using echo-planar T2* images with BOLD (blood oxygenation-level-dependent) contrast, and angled 30° with respect to the AC-PC line to minimize susceptibility artifacts in the orbitofrontal cortex (1). MR imaging settings were as follows: repetition time (TR) = 2,000 ms; echo time (TE) = 40 ms; slice thickness = 3 mm yielding a 64 × 64 × 32 matrix (3 mm × 3 mm × 3 mm); flip angle = 90°; FOV read = 220 mm; FOV phase = 100 mm, interleaved series order. High-resolution structural T1-weighted scans (1 mm × 1 mm × 1 mm) were acquired by using an MPRage sequence. Visual stimuli were presented by means of a mirror mounted on the MRI head coil, and responses were acquired via an MRI-safe button response pad (Neuroscan).

Computational Modeling. To characterize the relative contributions of reinforcement (RL) and belief-based learning to behavior, we considered three different models of learning: reinforcement learning, belief-based learning, and their hybrid, experience-weighted attraction (EWA). We first describe the hybrid model because it contains RL and belief learning models as special cases (2). First, denote s_i^k as strategy k for player i, $s_i(t)$ is the chosen strategy by player i at period t, and $s_{-i}(t)$ is the chosen strategy of the opponent at period t. Player i's expected reward, $V_i^k(t)$, for playing strategy s_i^k in period t is governed by three parameters and updates according to the following:

$$V_i^k(t) = \begin{cases} \dfrac{\phi_i \cdot N(t-1) \cdot V_i^k(t-1) + \pi_i(s_i^k, s_{-i}(t))}{N(t)}, & \text{if } s_i^k = s_i(t) \\ \dfrac{\phi_i \cdot N(t-1) \cdot V_i^k(t-1) + \delta_i \cdot \pi_i(s_i^k, s_{-i}(t))}{N(t)}, & \text{if } s_i^k \neq s_i(t), \end{cases} \quad [S1]$$

where parameter ϕ_i and function $N(t) = \rho_i \cdot N(t-1) + 1$ capture different aspects of the depreciation of $V_i^k(t)$. For example, if the player believes his opponent is a fast adaptor, he will have a small ϕ_i that depreciates past values faster. In contrast, ρ_i is the discount rate for the strength of past experience $N(t)$, and controls the influence of the out-of-game prior beliefs. If ρ_i is large, the out-of-game prior beliefs will wear off quickly. The third and most important parameter for our study, δ_i, is the weight between foregone payoffs and actual payoffs when updating values, and reflects one of the key insights of the hybrid model that belief learning is equivalent to a model whereby actions are reinforced by foregone payoffs in addition to received payoffs as in RL models. Thus, δ_i can be interpreted as a psychological inclination toward belief learning (2). That is, the hybrid model reduces to the RL model when $\delta_i = 0$, and the belief learning model when $\delta_i = 1$.

In belief learning, we also impose the restriction that the initial attractions are expected payoffs given some underlying probabilistic belief inference of the subject, that is, $V_i^k(0) = \sum_m q_{-i}^m(0) \times \pi_i(s_i^k, s_{-i}^m)$, where $q_{-i}^m(0)$ is player i's initial belief about the likelihood of his opponent adopting s_{-i}^m. Hence, $q_{-i}^m(0) \geq 0$ and $\sum_m q_{-i}^m(0) = 1$. The restriction ensures that in all of the trials that follow the belief learners update a probabilistic belief inference regarding the next move of the opponents rather than an unconstrained vector of fictive errors defined as the discrepancy between forgone payoffs and previous attraction values.

Behavioral Data Analysis. To calibrate the models given behavior of the subjects in the game, we estimated parameters of each model by using responses of subjects by maximizing the logistic log likelihood of the model predictions. To convert values into choices, we used a logit or softmax function to calculate the probability of player i playing strategy k in the next round, $p_i^k(t+1) = e^{\lambda_i \cdot V_i^k(t)} / \sum_{l=1}^{L} e^{\lambda_i \cdot V_i^l(t)}$, where λ_i is a measurement of sensitivity of subjects to difference in expected reward associated with the different actions.

Using these choice probabilities, we performed maximum likelihood estimation with a grid search over a large range of values for all free parameters in all estimations, because the likelihood function is not globally concave. Both pooled and individual-level estimations were performed. For pooled estimation, we aggregated observations conditional on the roles of the subjects and then fit the choice data by maximizing the log likelihood of the observed choices over rounds for subject $_i$. That is, $\sum_i \sum_t \log(p_i^{s_i(t)}(t))$. Although using pooled estimates is more robust in general, it removes the possible individual variation in learning, and will bias estimates due to heterogeneity (3). Therefore, we also performed estimation at the individual level. The primary challenge of individual estimation is the relatively small sample size compared with the number of free parameters. We approached this problem with two methods combined: (*i*) estimating a common set of initial attractions shared by all subjects with the same role, from the pooled first period of data, conditional on the role of the subject and (*ii*) self-tuning estimation as introduced in Ho et al. (4). As a robustness check, we also conducted individual level estimation with partially joint estimates across different roles, assuming each subject shares a subset of learning parameters (e.g., the decay rate of the initial belief) regardless of her role in the game. We found that the estimates to be robust across these different estimation strategies.

Conversion to Temporal Difference Form. To derive trial-by-trial predictors for use in neuroimaging analysis, we converted the

respective models above to a TD form whereby learning results from updating reward predictions through a prediction error. Choice probabilities and prediction errors on each trial were then generated by using the best-fit parameters derived from the behavioral data estimation. That is, we separated player i's expected reward, $V_i^k(t)$, for playing strategy s_i^k in period t into a reward prediction $V_i^k(t-1)$, and the prediction error that is the difference between the expected reward and obtained (foregone) reward $\pi_i(s_i^k, s_{-i}(t))$. In the hybrid model, the expected reward thus evolves according to:

$$V_{i,k}^{EWA}(t) = \begin{cases} V_{i,k}^{EWA}(t-1) + \frac{1}{N(t)}\left\{\pi_i(s_{i,k}, s_{-i}(t)) - V_{i,k}^{EWA}(t-1)\right\} & \text{if } s_{i,k} = s_i(t) \\ \underbrace{V_{i,k}^{EWA}(t-1)}_{\text{Reward Prediction}} + \frac{1}{N(t)}\underbrace{\left\{\delta_i \cdot \pi_i(s_{i,k}, s_{-i}(t)) - V_{i,k}^{EWA}(t-1)\right\}}_{\text{Prediction Error}} & \text{if } s_{i,k} \neq s_i(t). \end{cases} \quad [S2]$$

In contrast, RL updates by reinforcing only the chosen strategy, whereas belief learning updates by reinforcing all available strategies proportional to the possible rewards:

$$\text{RL}: V_{i,k}^{RL}(t) = \begin{cases} V_{i,k}^{RL}(t-1) + (1-\phi_i)\left\{\frac{1}{1-\phi_i}\pi(s_{i,k}, s_{-i}(t)) - V_{i,k}^{RL}(t-1)\right\} & \text{if } s_{i,k} = s_i(t) \\ \phi_i \cdot V_{i,k}^{RL}(t-1) & \text{if } s_{i,k} \neq s_i(t) \end{cases} \quad [S3]$$

$$\text{BB}: V_{i,k}^{BB}(t) = V_{i,k}^{BB}(t-1) + \frac{1}{N(t)}\left\{\pi(s_{i,k}, s_{-i}(t)) - V_{i,k}^{BB}(t-1)\right\}, \forall\, s_{i,k} \quad [S4]$$

fMRI Data Analysis. Image analysis was performed by using *SPM2* (Wellcome Department of Imaging Neuroscience, Institute of Neurology, London). Preprocessing included, in order: slice time correction (centered at *TR*/2), motion correction, coregistration, spatial normalization to the Montreal Neurological Institute (MNI) template, and spatial smoothing using an 8-mm Gaussian kernel (5). All images were also high-pass filtered in the temporal domain (width 128s) and autocorrelation of the hemodynamic responses was modeled as an $AR(1)$ process.

Analyses of fMRI time series were done by using standard random effects models (6), with reward prediction and prediction error values generated from the respective computational models calibrated on choices of subjects at the individual level. An event-related design was used where regressors were included for the decision and feedback events of the trials (Fig. 1). That is, for each subject, we constructed a (first level) general linear model (GLM) consisting of two events: an event at the time of decision, and one at the time of feedback. Regressors were constructed by using the trial-by-trial outputs from the TD form of the best-fitting individual parameter estimates. The decision event was associated with choice probabilities, which can be regarded as relative reward predictions controlled for time influence. The feedback event was associated with prediction errors for chosen actions. All analyses were performed on the feedback event data, except the expected reward region analysis (Fig. S3). The first eight rounds were excluded from the GLM analysis to allow initial values to stabilize. Regressors were convolved with the canonical hemodynamic response function and entered into a regression analysis against each subject's BOLD response data. The regression fits of each computational signal from each individual subject were then summed across their roles and then taken into random-effects group analysis.

SI Results

Comparison of Behavior Across Experimental Protocols. To measure the effectiveness of this protocol, we compared both aggregate choices and model estimates among (*i*) our neuroimaging subjects, (*ii*) our behavioral subjects, and (*iii*) Rapoport and Amaldoss's original experiment (7) (Table S1). Proportions of choices are similar, as are parameter estimates across the three different datasets. We found no evidence that the pool player protocol systematically affected behavior of players.

To further check the robustness of our pool player protocol, we compared behaviors in our strategic setting versus those in a matching but nonstrategic reward task. In the reward treatment, we replaced the human pool players with a computer algorithm. In contrast to the strategic treatment, subjects in the reward treatment were told to exceed a random hurdle determined by the computer to win the prize. Subjects were informed that they are playing against a computer algorithm. All other aspects of the instructions remained identical. In terms of the game display, the only difference was that in the reward treatment, the word "Opponent" was replaced with the word "Hurdle".

We found that learning in a reward setting is primarily RL-based. Using model-based estimates, we found that the hybrid δ parameter was significantly greater in the strategic treatment than the reward treatment ($P < 0.01$, two tailed). Visually, this difference can be illustrated through the transition matrices of the choices of players (Fig. S1). These matrices show how players switched their choices from one trial to the next and are generalizations of more traditional switch/stay measures (2). The diagonal elements indicate choices in which subjects stayed, whereas off-diagonals indicate switches.

The most striking features are the similarities between the transition matrices of the strategic treatment and the belief learning simulation (Fig. S1 *A* and *C*) and between reward treatment and RL simulation (Fig. S1 *B* and *D*). In particular, whereas players in the strategic treatment switched quite often, players in the reward treatment repeatedly played the same strategies, rarely switching between strategies from trial to trial. This behavior is apparent in that most of the mass of the transition matrix for the RL treatment and simulation is located along the diagonal (indicating stay trials) at investments of 1, 3, and 5 (Fig. S1*B*). At the aggregate level, the switch rate in the strategic treatment was 0.56 (exactly that of the Nash equilibrium prediction) versus 0.32 for the reward treatment. This finding is thus is consistent with the hypothesis that learning in the reward treatment is subserved primarily by reinforcement learning, which adapts more slowly.

Correlation of RL and Belief-Based Prediction Errors. Table S3 shows the correlation between the prediction errors associated with the three models under consideration. Crucially, we find that the correlation between RL and belief prediction errors is low (Pearson $\rho = 0.28$). The statistical separation between the model-generated learning signals indicates the potential to disentangle the unique contributions of the different types of learning signals. The correlation of reinforcement and belief-based learning with the hybrid model is not surprising, given the reinforcement and belief learning are nested models.

Orthogonality Tests on Robustness of Brain Activations. Although the correlation between RL and belief prediction errors was low, we nevertheless sought to investigate whether our reinforcement and belief prediction errors are robust to orthogonalization of the regressors. We verified that activations in response to RL and belief prediction errors remain after they are orthogonalized against each other (Fig. S2). The procedure is same as those described in ref. 8.

Expected Reward Regions. We found activity in ventromedial prefrontal cortex, extending to rACC and medial orbitofrontal cortex, to be correlated with the relative expected reward value of the chosen action (Fig. S3). The relative expected reward is defined as the probability generated from the different models for the chosen action at the time of response on a given trial. We used this notion to remove the possible time trend in the absolute expected reward values. This result is consistent with existing evidence on the role of orbital and adjacent medial prefrontal cortex in encoding predictions of future reward (9, 10).

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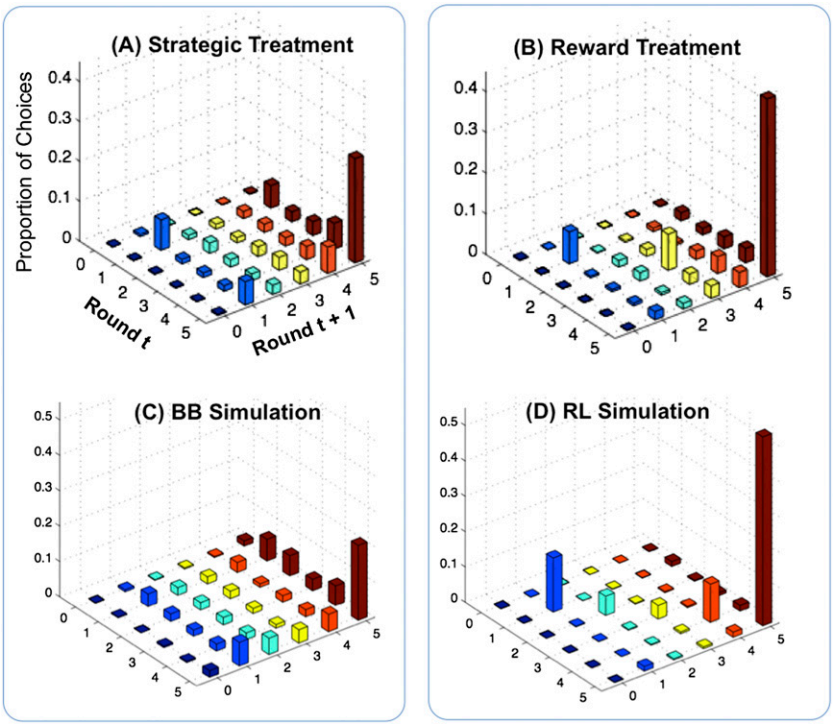


Fig. S1. Comparison of strategic and reward learning in Strong role. (*A* and *B*) Empirical frequency of transitions for strategic and reward treatments, respectively. (*C* and *D*) Transition matrices of simulations using belief and reinforcement learning models, respectively. Note behavior in strategic treatment is qualitatively more similar to the belief learning simulation, whereas reward treatment is more similar to the RL simulation.

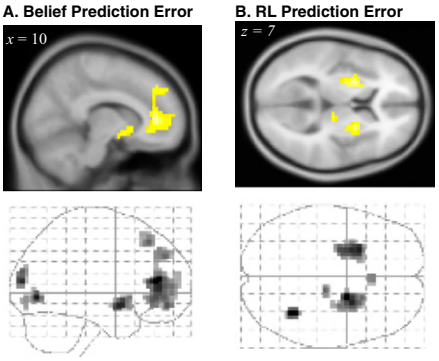


Fig. S2. Robustness check for orthogonalization between RL and belief learning prediction errors. (*A*) Belief learning prediction errors after orthogonalization against RL prediction errors. ($P < 0.001$, uncorrected, cluster size $k > 10$ voxels). (*B*) RL prediction errors after orthogonalization against belief learning prediction errors. ($P < 0.001$, uncorrected, cluster size $k > 10$ voxels).

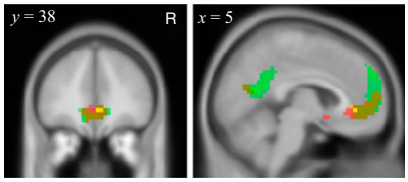


Fig. S3. Expected reward regions. Activity in ventromedial prefrontal cortex, extending to rACC and medial orbitofrontal cortex, is correlated with respect to relative expected reward value of the chosen action calculated under the hybrid (red), belief (yellow), and RL (green) models ($P < 0.005$ uncorrected, cluster size $k \geq 5$).

Table S1. Comparison of Nash equilibrium predictions and empirical distributions from (*i*) Rapoport and Amaldoss (1), (*ii*) our behavioral experiment, (*iii*) our neuroimaging experiment, and (*iv*) a reward learning control session

			Empirical distributions			
Role	Investment	Equilibrium prediction, %	Matrix form, %	Behavioral session, %	Neuroimaging session, %	Reward learning, %
Strong	0	0	1	0	1	1
	1	20	17	14	18	11
	2	0	5	6	10	6
	3	20	9	13	11	16
	4	0	13	25	16	11
	5	60	55	43	45	54
Weak	0	60	55	49	49	30
	1	0	3	3	4	12
	2	20	6	10	7	18
	3	0	14	10	14	8
	4	20	22	28	27	32

Empirical distribution is proportion of all players' choices over all rounds.

Table S2. Median individual level estimates

Model	δ	ϕ	λ
Reinforcement	0*	0.94 (0.86, 0.96)	0.04 (0.02, 0.07)
Belief-based	1*	0.95 (0.83, 0.98)	0.60 (0.23, 2.11)
Hybrid	0.46 (0.29, 0.69)	0.71 (0.53, 0.81)	0.51 (0.32, 0.70)

Parentheses contain first and third quartile of empirical distribution.
*Parameters constrained by model.

Table S3. Correlation coefficient between the prediction errors from different learning models

Model	Reinforcement	Belief-based	Hybrid
Reinforcement	—	(0.16)	(0.10)
Belief-based	0.28	—	(0.18)
Hybrid	0.63	0.40	—

Parentheses contain SDs for the correlation coefficients.